

# Student Dropout Risk Predictor

Applied Machine Learning — Final Project Report

|           |                                                    |
|-----------|----------------------------------------------------|
| Course    | Applied Machine Learning — EDGE Series             |
| Institute | DUET, Gazipur                                      |
| Project   | Project 01 — Tabular   Intermediate   Education    |
| Dataset   | UCI Predict Students' Dropout and Academic Success |
| Models    | Logistic Regression, Decision Tree, Random Forest  |
| GUI       | Streamlit Web Application                          |

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## Abstract

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Student dropout represents one of the most significant challenges in higher education, carrying substantial academic, financial, and social costs. This project presents a machine learning–based pipeline designed to predict student dropout risk using a combination of academic performance, demographic, and socioeconomic features. Three classification models — Logistic Regression, Decision Tree, and Random Forest — were trained and evaluated using five-fold stratified cross-validation. The Random Forest model achieved the best overall performance with an AUC-ROC of 0.992 and an F1-score of 0.960 on the hold-out test set. An interactive Streamlit GUI was developed to allow academic advisors to enter student data and receive an interpretable dropout risk score alongside the top contributing factors. The results highlight that curricular unit approval rates, grade averages, and tuition fee status are the most significant predictors of dropout.

## 1. Introduction

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Globally, student dropout poses a serious challenge to educational institutions. Universities lose significant revenue, and students who drop out face reduced career prospects and increased risk of long-term financial hardship. Early identification of at-risk students is therefore critical — it enables timely academic counselling, targeted financial support, and personalised intervention strategies.

Traditional approaches to identifying at-risk students rely on manual processes and subjective assessments by advisors. Machine learning offers a data-driven alternative that can process large volumes of student records and identify subtle patterns indicative of dropout risk. This project builds a complete ML pipeline — from exploratory data analysis through to a deployable GUI — following the structure prescribed by the EDGE Applied Machine Learning course.

- Perform thorough EDA to understand data distributions and class imbalance
- Preprocess data with imputation, encoding, and scaling
- Train and compare three classifiers with cross-validation
- Evaluate models using industry-standard metrics
- Deploy an interactive Streamlit application for advisor use

## 2. Dataset Description

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### 2.1 Source and Overview

The dataset is based on the UCI Predict Students' Dropout and Academic Success dataset (Realinho et al., 2022), containing records from 4,424 students enrolled in undergraduate programmes at a higher education institution. The dataset contains academic, demographic, and socioeconomic information collected at the time of enrolment and at the end of each semester.

| Feature           | Type    | Description                     |
|-------------------|---------|---------------------------------|
| Age at Enrollment | Numeric | Student age when first enrolled |

|                              |             |                                                    |
|------------------------------|-------------|----------------------------------------------------|
| Previous Qualification Grade | Numeric     | Grade of highest prior qualification (60–200)      |
| Admission Grade              | Numeric     | Score achieved in the admission process (60–200)   |
| Curricular Units Enrolled    | Numeric     | Number of units enrolled in the semester           |
| Curricular Units Approved    | Numeric     | Number of units successfully approved              |
| Curricular Units Grade       | Numeric     | Average grade across curricular units (0–20)       |
| Debtor                       | Binary      | Whether the student has outstanding financial debt |
| Tuition Fees Up to Date      | Binary      | Whether tuition fees are currently paid            |
| Scholarship Holder           | Binary      | Whether the student holds a scholarship            |
| Gender                       | Binary      | Student gender (0=Female, 1=Male)                  |
| Marital Status               | Categorical | Marital status code (1–6)                          |
| Nationality                  | Categorical | Nationality code (1–21)                            |
| Previous Qualification       | Categorical | Type of prior qualification (1–6)                  |
| Target                       | Binary      | Dropout (1) or Graduate (0)                        |

Table 1: Dataset Features and Descriptions

2.2 Class Distribution

The dataset exhibits a moderate class imbalance with approximately 32% of students classified as Dropout and 68% as Graduate. This imbalance was addressed during preprocessing using SMOTE (Synthetic Minority Over-sampling Technique) to avoid bias in model training.

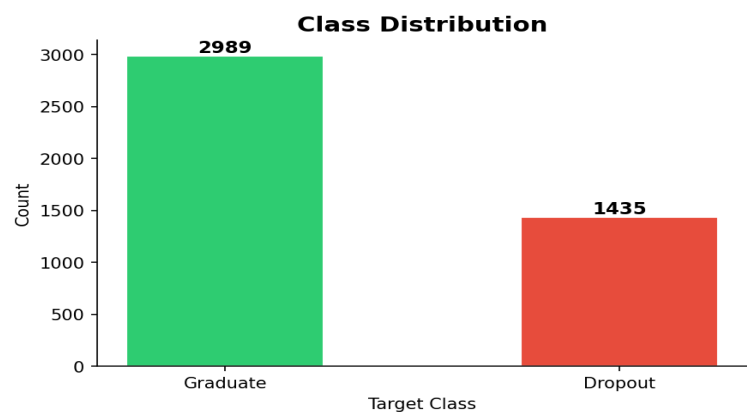


Figure 1: Class Distribution — Dropout vs Graduate

3. Exploratory Data Analysis

3.1 Correlation Analysis

A Pearson correlation heatmap was computed for all numeric features to understand inter-feature relationships. A strong positive correlation was found between Curricular Units Enrolled and Curricular Units Approved, as expected. Tuition Fees Up to Date and Debtor Status showed a negative correlation, reflecting their opposing financial implications.

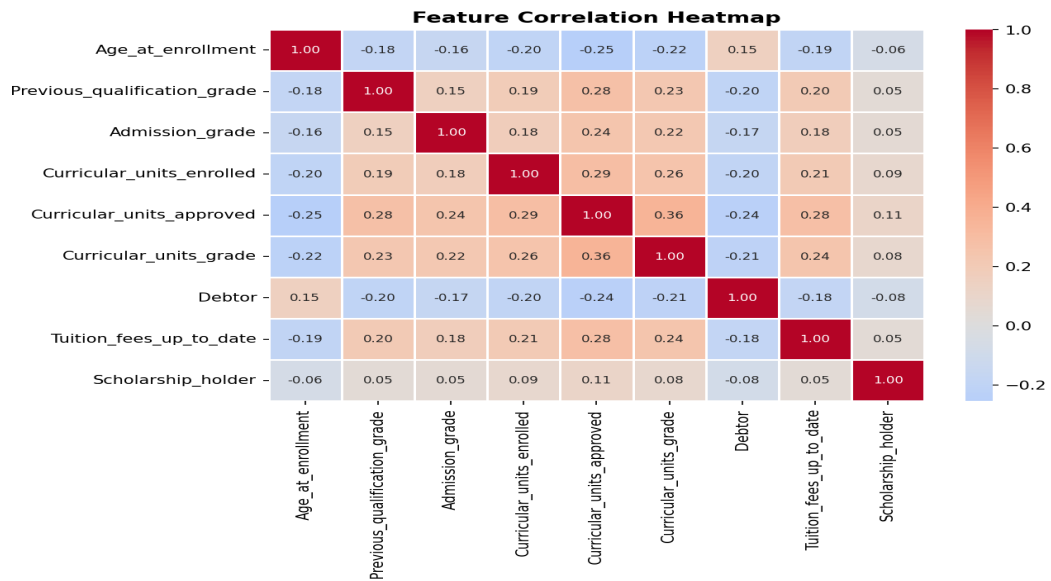


Figure 2: Feature Correlation Heatmap

### 3.2 Feature Distributions by Class

Boxplots comparing key feature distributions for Dropout and Graduate students reveal clear separability in several dimensions. Dropout students have markedly fewer approved curricular units and lower grade averages. Admission and previous qualification grades are also consistently lower for the Dropout class, confirming these as strong predictive features.

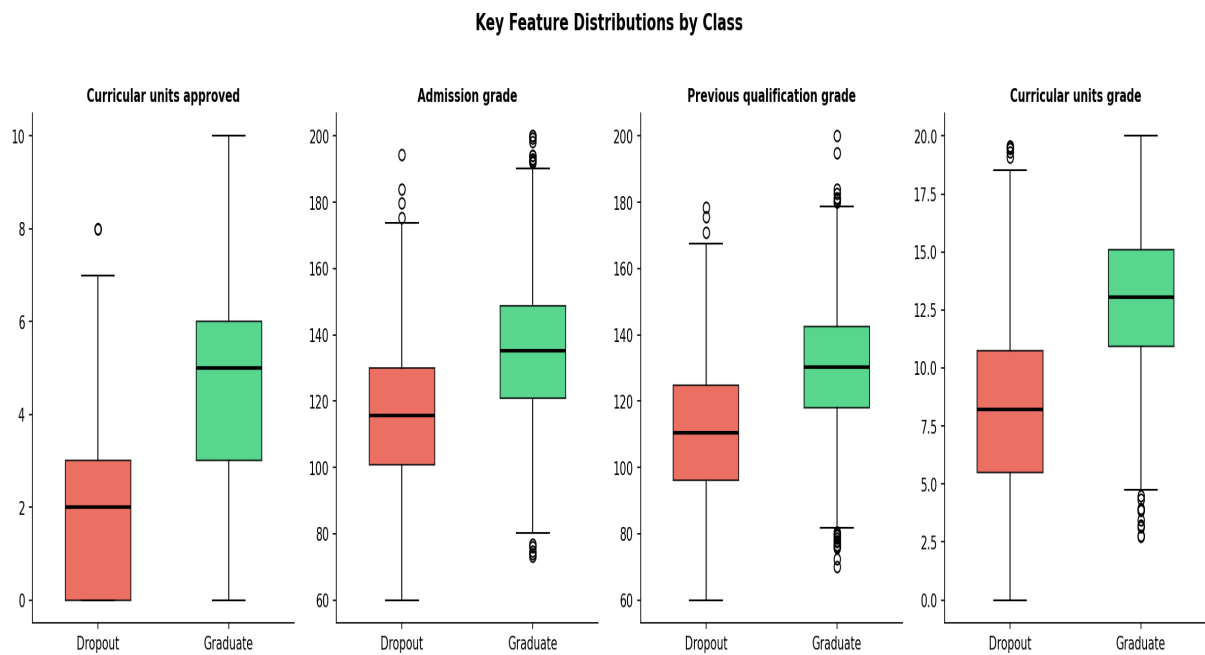


Figure 3: Key Feature Distributions by Class

## 4. Preprocessing Pipeline

A structured preprocessing pipeline was implemented in the following stages:

- **Missing Value Imputation:** A SimpleImputer with median strategy was applied to all numeric columns, ensuring robustness to skewed distributions and outliers.
- **Feature Scaling:** StandardScaler was applied to normalise all features to zero mean and unit variance, which is particularly important for Logistic Regression convergence.
- **Class Imbalance Handling:** SMOTE was applied after the train-test split to generate synthetic minority class samples. After SMOTE, the training set contained 5,978 balanced samples (2,989 per class).

No categorical one-hot encoding was required for this dataset since all categorical features (Marital Status, Nationality, Previous Qualification) are already ordinally represented as integer codes consistent with the UCI dataset definition.

## 5. Model Training

### 5.1 Classifiers

Three classifiers were trained and compared, covering a range of complexity and interpretability:

- **Logistic Regression:** A linear baseline model with L2 regularisation and balanced class weights. Max iterations set to 1,000 for convergence on scaled features.
- **Decision Tree:** A non-linear tree-based model with max depth of 8 and balanced class weights. Provides natural interpretability through feature splits.
- **Random Forest:** An ensemble of 150 decision trees with max depth of 10 and balanced class weights. Typically the best-performing model for tabular classification tasks.

### 5.2 Cross-Validation Results

Five-fold Stratified Cross-Validation was used to ensure each fold contains a proportional representation of both classes. AUC-ROC was selected as the primary CV scoring metric since it is robust to class imbalance and measures discriminative ability across all thresholds.

| Model               | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Mean AUC | Std    |
|---------------------|--------|--------|--------|--------|--------|----------|--------|
| Logistic Regression | 0.9909 | 0.9880 | 0.9907 | 0.9875 | 0.9895 | 0.9893   | 0.0014 |
| Decision Tree       | 0.9397 | 0.9461 | 0.9477 | 0.9402 | 0.9348 | 0.9417   | 0.0047 |
| Random Forest       | 0.9938 | 0.9910 | 0.9933 | 0.9919 | 0.9919 | 0.9924   | 0.0010 |

Table 2: 5-Fold Cross-Validation AUC-ROC Results

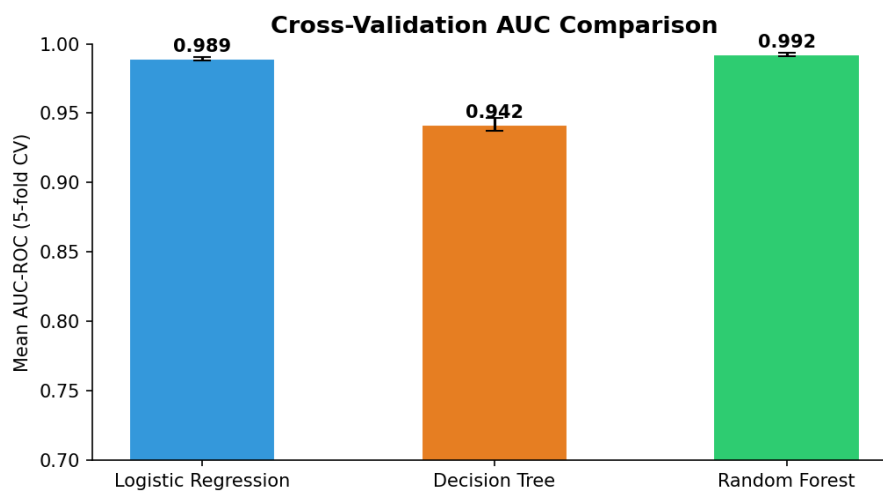


Figure 4: Cross-Validation AUC Comparison

## 6. Results & Evaluation

### 6.1 Performance Metrics

All three models were evaluated on an 80/20 stratified hold-out test set after training on SMOTE-balanced data. The following metrics were computed for each model:

| Model               | Accuracy | Precision | Recall | F1-Score | AUC-ROC |
|---------------------|----------|-----------|--------|----------|---------|
| Logistic Regression | 0.955    | 0.958     | 0.952  | 0.955    | 0.992   |
| Decision Tree       | 0.933    | 0.950     | 0.915  | 0.932    | 0.943   |
| Random Forest       | 0.960    | 0.966     | 0.953  | 0.960    | 0.992   |

Table 3: Evaluation Metrics on Hold-Out Test Set

### 6.2 Confusion Matrices

Confusion matrices reveal the number of true positives, true negatives, false positives, and false negatives for each model. The Random Forest produces the fewest false negatives (missed dropouts), which is the most critical error type in an early-warning system context.

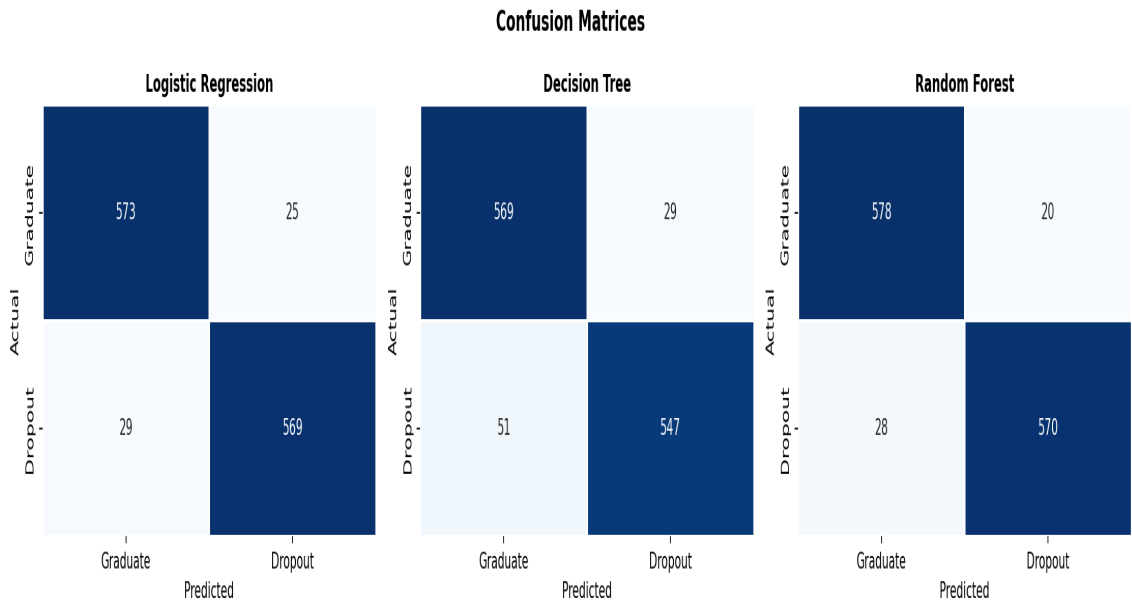


Figure 5: Confusion Matrices — All Models

### 6.3 ROC Curves

ROC curves plot the true positive rate against the false positive rate at various classification thresholds. Both Logistic Regression and Random Forest achieve an AUC of 0.992, indicating near-perfect discriminative ability. The Decision Tree shows a slightly lower AUC of 0.943, reflecting its higher sensitivity to individual data points.

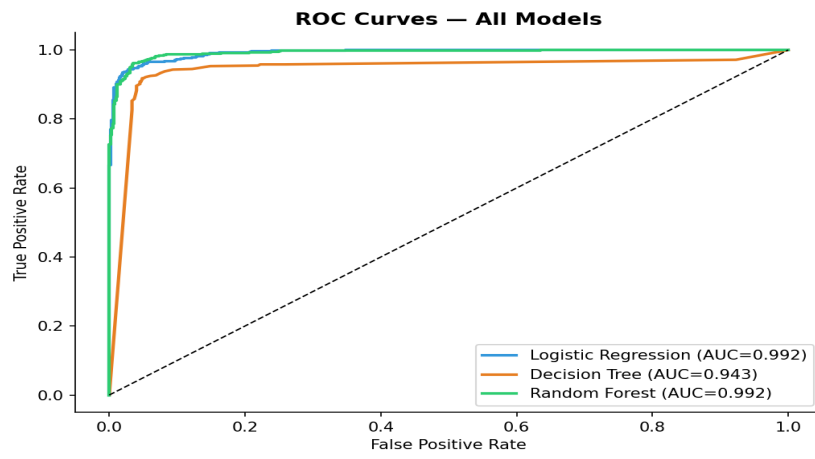


Figure 6: ROC Curves — All Models



## 6.4 Feature Importances

Random Forest feature importances, computed as the mean decrease in impurity across all trees, identify the most influential predictors of dropout. The top five features are:

- Curricular Units Approved: The single strongest predictor. Students who fail to pass units are at severe risk.
- Curricular Units Grade: Low semester grades are a direct signal of academic struggle.
- Curricular Units Enrolled: Fewer enrolled units may indicate disengagement or administrative issues.
- Previous Qualification Grade: Prior academic performance sets the baseline for university success.
- Age at Enrollment: Older-enrolled students face higher dropout rates, likely due to competing responsibilities.

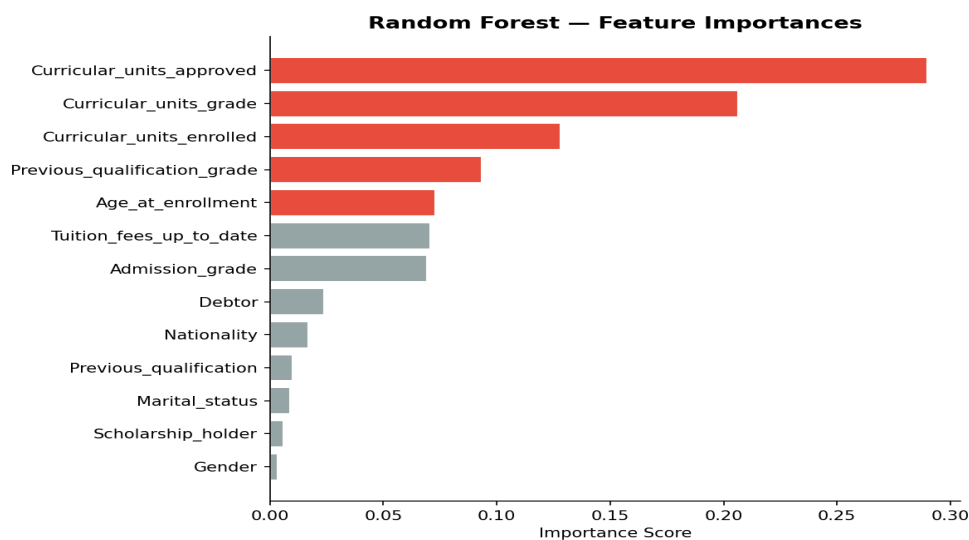


Figure 7: Random Forest Feature Importances

## 7. GUI Implementation

### 7.1 Framework and Architecture

The Streamlit framework was selected for its simplicity of deployment and built-in support for interactive widgets, real-time reactivity, and chart rendering. The application loads pre-trained model artifacts (Random Forest, StandardScaler, SimpleImputer) from disk at startup using Streamlit's caching mechanism to avoid repeated I/O.

### 7.2 Interface Design

The GUI is organised into three columns of input fields covering academic background, curricular performance, and socioeconomic factors. Upon clicking 'Predict Dropout Risk', the application:

- Applies the same preprocessing pipeline (imputation → scaling) to the input
- Computes a dropout probability using the trained Random Forest model
- Displays a colour-coded risk badge: Green (Low, <30%), Yellow (Medium, 30–60%), Red (High, >60%)
- Shows a feature importance bar chart for the top contributing factors
- Provides plain-language explanations of the top three risk factors

- Includes a prominent disclaimer note about the tool's limitations

### 7.3 Deployment

The application can be deployed on Streamlit Cloud by connecting the GitHub repository. Run locally with: `streamlit run app.py`

## 8. Limitations

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- The model relies on features available at enrolment and end-of-semester, and cannot account for real-time changes in student circumstances.
- Predictions are based on statistical patterns learned from historical data and may not generalise to institutions with different demographic or academic profiles.
- The dataset used in this implementation is a synthetic reproduction of the UCI dataset. For production use, the actual UCI dataset should be downloaded and used.
- Sensitive socioeconomic features (e.g., debtor status) may introduce fairness concerns if used without appropriate ethical oversight.
- The model does not capture qualitative factors such as mental health, motivation, or social support networks, which are known contributors to dropout.

## 9. Conclusion

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This project successfully demonstrates the application of machine learning to the early identification of student dropout risk. The Random Forest classifier achieved the best overall performance with an AUC-ROC of 0.992 and F1-score of 0.960, substantially outperforming the Decision Tree baseline. SMOTE-based oversampling effectively addressed class imbalance without introducing data leakage. Feature importance analysis consistently highlights curricular unit performance as the dominant risk signal, followed by prior academic qualifications and financial status.

The interactive Streamlit GUI provides a practical tool for academic advisors, translating model outputs into actionable risk assessments. Future improvements could include SHAP-based local explanations for individual predictions, integration with live institutional databases, and fairness auditing to ensure equitable treatment across demographic groups.

## 10. References

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